

Vidvathama Ramesh

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Research Interests

Visual navigation and embodied perception; dexterous manipulation; 3D feedforward scene reconstruction; NLP/LLM systems (routing, evaluation, safety); decision theory and strategic reasoning applied to multi-agent and RL systems.

Education

International Institute of Information Technology, Hyderabad (IIIT-H) **July 2024 – Present**
B.Tech in Electronics and Communication Engineering; CGPA: 9.01/10.0 (9.1+ in the last 3 semesters); Expected May 2028
Hyderabad, India

- **Relevant Coursework:** Statistical Methods in AI, Natural Language Processing, Processor Architecture, Digital Systems and Logic, VLSI Design, Signal Processing, Information Theory, Probability and Random Processes, Systems Thinking, Functional Analysis, Data Structures and Algorithms, UAV Design (EC4.402).

Research Experience

Research Intern — TCS Research **May 2026 – July 2026**
Mentor: Jeripothula Prudviraj *Hyderabad, India*

- Architected deep neural networks that extract high-level aesthetic and stylistic features as state representations for autonomous RL agents, optimizing policy-gradient reward functions for iterative generative decision-making.
- Engineered a human-AI co-creation framework integrating computer vision and RL for real-time agents that adapt to human creative intent; preparing submissions to **NeurIPS 2026** (Creative AI Track: Agency) [submitted] and **ACM CHI 2026**.

Undergraduate Researcher — Robotics Research Center (RRC), IIIT-H **May 2025 – Present**
Supervisor: Prof. Sourav Garg *Hyderabad, India*

- Integrating Vision-Language Actions/Models (VLAs/VLMs) with topological graph structuring for robust visual navigation in unstructured environments.
- Developing 3D-vision + RL models for dexterous manipulation on a LEAP Hand, with tactile simulation via TacSL in Isaac Lab.
- Currently benchmarking feedforward 3D mapping models (DUST3R/MASt3R/CUT3R lineage) against existing visual navigation pipelines as part of an ongoing sprint targeting an **ICRA 2026** submission.

Research Intern — Indian Institute of Technology Bombay (IIT B) **May 2025 – Aug 2025**
Supervisor: Prof. Vikram Bansal *Mumbai, India*

- Implemented a TransCrowd Transformer pipeline for high-density crowd counting, integrating spatio-temporal features to improve density estimation accuracy across large-scale datasets.

Manuscripts in Preparation

- **Adaptive Semantic Pipeline for LLM Routing** — ACL course paper (INLP), with full evaluation scripts, pipeline traces, and interactive demo.
- **CanvasMind: Multi-Agent Inference-Time RL for Creative AI** — ACM CHI in progress, NeurIPS: Creative AI Track (submitted), co-authored.
- **3D Feedforward Mapping for Visual Navigation** — targeting ICRA 2026, Robotics Research Center, IIIT-H.
- **HiGrEC: Hierarchical Classification for Political Question Evasion Detection** — SemEval-2026 Task 6 (CLARITY) system paper, team Nier_ANLP.

Technical Projects

Adaptive Semantic Pipeline (ASP) for LLM Routing **Jan. 2026**
DistilBERT, PyTorch, LightGBM, spaCy

- Architected an 8-stage LLM routing pipeline decomposing prompts into a semantic DAG (spaCy, SentenceTransformer, constituency parsing) and routing sub-tasks to strong/weak models via a 27-feature LightGBM router with BFS strong-inheritance (85.66% accuracy, 88.04% F1), cutting inference cost $\sim 49\%$ at 131 ms mean latency.
- Fine-tuned a dual-head DistilBERT (11-class intent + complexity regression; 83.32% accuracy, 85.32% F1) and built an async DAG execution engine with speculative execution, semantic caching, and a context blackboard (60–80% token savings).
- Layered a 3-stage safety, NLI-verification, and LLM-as-a-Judge stack on top of the routing and aggregation pipeline.

CanvasMind — Multi-Agent Inference-Time RL for Creative AI **May 2026**
FastAPI, Azure OpenAI, Python

- Built a multi-agent creative-AI system in which generative agents co-paint a shared canvas under an inference-time RL layer using best-of-N search, a batched reward model, a persistent UCB1 bandit, and reward-triggered Reflexion self-correction.
- Modeled agent interactions with misaligned per-agent rewards as a general-sum game; applied Shapley-value, empowerment, and Goodhart diagnostics to identify and correct 4 RL defects, lifting composite quality from -0.05 to $+1.66$ at equal

compute across a 4,000-session ablation.

PolyQuery (PolyRA) — Multi-Model Relational Algebra Engine

Nov. 2025

C++17, Systems, Concurrency

- Engineered a high-performance, integer-optimized C++17 RDBMS unifying relational algebra, graph processing, and high-dimensional vector search over a custom block-based buffer manager with external-memory algorithms (external merge sort, hashing, B-Tree).
- Implemented multi-threaded transactions with Wait-Die concurrency control and page-granular shared/exclusive locking (deadlock-free), disk-backed BFS path queries, and KNN search (Euclidean/Cosine) via a bounded min-heap under a fixed 10-page memory ceiling.

32-bit RISC-V Processor

May 2025

Verilog, iVerilog, Python, RISC-V Assembly

- Designed a 32-bit RISC-V processor in Verilog, scaling from a single-cycle sequential datapath to a 5-stage IF/ID/EX/MEM/WB pipeline, including PC, register file, instruction/data memories, ALU/control, and immediate-generation units for R-, I-, S-, and B-type instructions.
- Engineered forwarding and hazard-detection units for data, double-data, load-store, control, and load-use hazards using forwarding, stalls, and branch flushes; built a Python assembler and iVerilog testbenches validating Fibonacci, linear search, vector addition, and summation programs.

POSIX-Core C Shell

Aug 2025

C, Unix Syscalls, Signals, Valgrind

- Developed a modular `execvp`-based Unix shell with foreground/background execution, unlimited pipelines, full I/O redirection, and `.myshrc` alias resolution; verified leak-free execution with Valgrind.
- Implemented 12+ built-ins (`hop`, `reveal`, `seek`, `log`, `proclore`, `activities`, `fg/bg`, `neonate`, and an `iMan` TCP man-page fetcher) with robust SIGINT/SIGTSTP/SIGCHLD handling and asynchronous child reaping.

Additional Research & Coursework Projects

HiGrEC — Hierarchical Classification for Political Question Evasion

Jan 2026

ANLP Course Project, Team Nier_ ANLP

- Selected SemEval-2026 Task 6 (CLARITY) as the shared task; designed a hierarchical classification system over the QEvasion benchmark, formalized in a finalized 2-page ACL-format proposal.

Gyrocopter (Autogyro) Design and Testing

March 2026

UAV Design (EC4.402), ArduPilot, XFLR5

- Assembled a free-spinning-rotor autogyro prototype (fixed-wing fuselage, forward-thrust BLDC propulsion); configured ArduPilot/ArduPlane firmware for stabilization assistance.
- Performed aerodynamic analysis (blade element/airfoil selection) in XFLR5 and derived flight dynamics calculations for rotor sizing and stability.

Quadrotor Trajectory Planning and ROS Simulation

March 2026

Mobile Robotics Coursework, MATLAB, Python, ROS

- Derived acceleration-to-attitude control mappings (thrust, roll, pitch commands) for multi-waypoint quadrotor navigation and simulated the resulting trajectories in ROS.

Sequence Modeling: Cipher Decryption and Masked Language Modeling

Jan 2026

INLP Coursework, PyTorch, HuggingFace, WandB

- Built custom RNN/LSTM models (no `nn.RNN/nn.LSTM`) for character-level cipher decryption and SSM/BiLSTM models for masked language modeling; logged experiments to WandB and published models to HuggingFace (Vidvathama).

Transformer, CNN, and Optimization Study

Jan 2026

SMAI Coursework, PyTorch

- Implemented a decoder-only Transformer comparing Pre-Norm vs. Post-Norm configurations, a CNN with Grad-CAM interpretability, and an optimizer/normalization ablation study on CIFAR-10.

Technical Skills

Languages: C, C++, Python, Verilog, RISC-V Assembly, x86 Assembly, JavaScript, SQL, Bash, MATLAB, L^AT_EX

Systems: Operating Systems, Linux/Unix, POSIX, System Calls, Signals, Concurrency, Parallel Computing, CPU Architecture, Digital Systems Design

ML & Frameworks: PyTorch, TensorFlow, Scikit-Learn, HuggingFace Transformers, LightGBM, OpenCV, NumPy, Pandas, spaCy, FastAPI, Flask, LangGraph, ROS 2, Isaac Gym/Isaac Lab

Web & App: Next.js, React, Node.js

Tools & Databases: Git, Docker, Valgrind, iVerilog, PostgreSQL, MySQL, Azure OpenAI

Achievements & Honors

- **Hackathon Winner (FinTech/Banking):** 1st Prize, SUD Life Bank Hackathon — intelligent insurance platform.
- **Lloyds Banking Agentic AI Hackathon:** Recognized entry; separately, Top 10 Finalist at the Lloyds IP&I Hackathon (2025) for a voice-first IVR replacement using RAG (Whisper, GPT-4o, FAISS) to ground responses in policy documents and auto-file tickets.
- **Dean's List:** Awarded for Academic Excellence — Top 5% of batch (Spring UG2) and Top 10% of batch (Monsoon UG3).